

Carl Monnaert

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EDUCATION

Imperial College London – London, United Kingdom 2026 - 2027

MSc in Statistics (Statistical Finance)

Relevant Coursework: Statistical Inference - Stochastic Processes - Time Series Analysis - Deep Learning

Télécom Paris - Institut Polytechnique de Paris – Palaiseau, France 2024 - 2027

MSc in Engineering - Diplôme d'Ingénieur (Applied Mathematics & Computer Science) **GPA: 3.9/4.0**

Relevant Coursework: Stochastic Calculus - Time Series - Optimization - Machine Learning

Lycée Roosevelt - "Preparatory Class" – Reims, France 2022 - 2024

Two-year undergraduate program (Mathematics & Computer Science)

Relevant Coursework: Probability Theory - Linear algebra - NP completeness - C/OCaml Programm

WORK EXPERIENCE

Quantitative Research Intern – Edo Theory (Paris, France) Jul 2026 - Aug 2026

- Investigated whether weekend cryptocurrency signals contain predictive information for Monday equity returns using five years of 30-minute cross-asset data through statistical significance testing

- Extended the approach to intra-week overnight cryptocurrency signals.

- Applied Kalman filtering to model time-varying relationships between cryptocurrency -signals and subsequent equity returns.

- Developed a Python backtesting framework incorporating transaction costs and systematic parameter search, utilizing walk-forward validation and out-of-sample testing to evaluate signal robustness and avoid look-ahead bias.

Math & CS Teacher – Masteur & Chôten Academie (Paris, France) Dec 2024 - Current

Delivered math and CS holiday and private courses to high-school and preparatory class students.

RELEVANT PROJECTS & CHALLENGES

Data Challenge — Qube Research & Technologies Oct 2025 - Jan 2026

Overall Survival Prediction for patients diagnosed with Myeloid Leukemia.

Built survival-analysis pipeline using Random Survival Forests for right-censored clinical data, integrating genomic features and risk factors; achieved C-index 0.761, ranking 138 out of 747 teams.

Carbon Cycle Simulation — Academic Project Feb 2026 - Apr 2026

Robustness testing and integration scheme benchmarking for nonlinear dynamical systems.

- Engineered a Python framework to solve an 8-state nonlinear ODE system.

- Implemented and benchmarked Runge-Kutta variants (Euler, Heun, RK4) and Adams-Moulton 3.

- Validated numerical robustness by injecting i.i.d. Gaussian noise and initial condition perturbations across trajectories to empirically bound worst-case error ratios and derive the system's stability constant.

- Conducted convergence and local truncation error analyses, verifying theoretical error bounds.

SKILLS

Quantitative Stochastic Calculus, Time Series, Machine Learning, Monte Carlo Simulations, Probability Theory, Statistical Inference, Optimization, Numerical Methods, Bayesian Statistics.

Technical Python (Pandas, NumPy, Scikit-Learn, PyTorch), OCaml, C, SQL, Git, LaTeX.

Linguistic Fluent in French and English, Conversational Proficiency in Spanish.

Activities Poker, Chess (1450+ rapid), Muay Thai.